

CAN Bus Message Analyzer

Module: Introduction to Automotive Electronics

1. Introduction

Modern vehicles use **electronic control units (ECUs)** to monitor and control different vehicle functions such as engine performance, braking, and safety systems. For communication between these ECUs, the **Controller Area Network (CAN) bus** protocol is widely used.

This program simulates a **CAN Bus Message Analyzer** using an **AUTOSAR-like layered architecture**. It demonstrates how CAN messages are processed from low-level access to application logic.

2. CAN Bus Communication Overview

The **CAN bus** is a serial communication protocol designed specifically for automotive applications. It allows multiple ECUs to communicate over a single network without a central controller.

Each CAN message consists of:

- **CAN ID** – Identifies the message and its priority
- **Data Field** – Contains actual sensor or status information
- **Control and CRC fields** – Ensure reliable data transmission

3. Simulated CAN Log File

A **simulated CAN log file** is used instead of live vehicle data. The log file contains recorded CAN messages, where each message represents sensor data transmitted by an ECU.

Using a log file helps in:

- Understanding CAN message structure

- Testing decoding logic without real hardware
- Simulating real-time vehicle behavior

4. Decoding CAN IDs and Signals

Each CAN message is identified using a **CAN ID**. The CAN ID is mapped to a specific vehicle parameter such as:

- Vehicle Speed
- Engine RPM
- Coolant Temperature

Decoding involves:

- Reading the CAN ID from the log file
- Identifying the corresponding signal
- Converting raw data bytes into meaningful physical values

5. Vehicle Parameters Monitored

5.1 Vehicle Speed

Vehicle speed indicates how fast the vehicle is moving and is usually calculated from wheel speed or speed sensors. Monitoring speed is essential for driver assistance and safety systems.

5.2 Engine RPM

Engine Revolutions Per Minute (RPM) represents the speed of the engine crankshaft. It is a critical parameter for engine control, fuel injection, and performance monitoring.

5.3 Coolant Temperature

Coolant temperature reflects the thermal condition of the engine. Maintaining the temperature within safe limits is necessary to avoid engine damage.

6. Temperature Safety Monitoring

A **safe temperature threshold** is defined in the system.

If the coolant temperature exceeds this threshold:

- The system raises a **warning message**
- This simulates an engine overheating alert

7. Console-Based CAN Monitoring Tool

The final output of **console-based CAN monitoring tool** that:

- Reads CAN messages from a log file
- Decodes CAN IDs into meaningful signals
- Displays vehicle parameters in real time
- Generates a warning when unsafe temperature conditions are detected

8. Conclusion

This provides practical exposure to:

- CAN bus communication fundamentals
- Decoding vehicle signals from CAN messages
- Monitoring critical vehicle parameters
- Implementing basic safety checks

TRACEBILITY MATRIX:

Req ID	Requirement	Code Function	Test Case	Status
R1	Read log file	<code>read_can_log()</code>	File exists/missing	✓
R2	Decode ID→signal	<code>decode_can_message()</code>	0x100→Speed	✓
R3	Display Speed/RPM/Temp	<code>application_*</code>	55km/h, 3000rpm, 85°C	✓
R4	Temp warning >100°C	<code>application_coolant_temp()</code>	105°C→WARNING	✓
R5	RTE routing	<code>rte_transmit()</code>	ID→correct function	✓